“I Trust It When I Understand It”: Understanding How Everyday Users Negotiate Trust, Control, Safety, and Communication in Semi-Autonomous Vehicles

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Abstract

Semi-Autonomous vehicles (SAVs) are increasingly being adopted by everyday users, yet little qualitative research exists on examining how these users negotiate trust, control, and communication with these systems. Through semi-structured interviews with two Tesla drivers, we conducted an inductive thematic analysis to understand how users evaluate SAV capabilities and make sense of semi-autonomous driving features, interpret system behavior, & adjust their trust in SAV automation. Through reflexive thematic analysis, three themes emerged. First, users judge SAV competence based on comparisons to human perception capabilities. Second, users conceptualize driving as a negotiated partnership in which control of the vehicle shifts between the human and the SAV, supported by predictable override mechanisms. Third, interaction and communication modalities especially visual displays, shape cognitive load, emotional comfort, and perceived system transparency. These findings offer practical guidance for designing SAV systems that better align with user expectations, cognitive needs, and everyday driving realities.

CCS Concepts

• **Human-centered computing** → **User interface design**; **HCI theory, concepts and models**; *Interaction design theory, concepts and paradigms*; • **Social and professional topics** → *User-centered design.*

Keywords

Human-automation trust; qualitative interviews; shared control; vehicle communication

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1 Introduction

Semi-Autonomous Vehicle (SAV) technology innovations have grown rapidly in the past decade and with this growth, has come a shift in public expectations around driving, safety, convenience, and human–machine collaboration [1]. While much of this improvements focus on aspects like technical performance, predictive accuracy, and societal adoption patterns, not much is understood about how everyday users make sense of SAV behavior, negotiate control, and develop confidence or hesitation when interacting with semi-autonomous systems [2]. This is a gap that needs to be resolved because as SAVs are increasingly being integrated into consumer markets, understanding how users form judgments about SAV competence, trustworthiness, control dynamics and communication modalities, becomes necessary in designing these autonomous vehicles to be safer, more usable and more trustworthy. Previous research has shown that trust in automation is influenced by many factors including system transparency, perceived reliability, communication modalities, and the user’s mental model of the system’s capabilities [4]. Similarly, human factors and HCI studies have emphasized that users prefer automation that augments their capabilities without making them feel incapacitated [7]. While these studies are important and relevant, research into the lived experiences of everyday SAV users especially those interacting with commercial vehicles which have been equipped with semi-autonomous features like Tesla, remain underexplored, and there is little on in-depth accounts from real SAV drivers navigating real-world contexts. Thus, there is still a lack of deeper qualitative insights into how everyday users interpret safety features, understand system boundaries, negotiate control handoffs, and evaluate communication modalities. Without these, it becomes difficult to design SAV systems that truly align with users’ cognitive, emotional, and situational needs.

To address this gap, our study uses a reflexive thematic analysis approach [3] to investigate how real SAV users describe their experiences with SAV control, system communication, and how they build their trust in the SAV automation. Through semi-structured interviews with two SAV drivers who regularly use semi-autonomous features, this work provides a deeper look into how trust is calibrated, how control is shared, and how communication influences user comfort and situational awareness.

Although SAV research often emphasizes technical performance, far less is known about how everyday users make sense of the system’s behaviors, especially during moments that require lightning-speed decision-making or careful interpretation of the SAV’s intentions. Because SAVs are entering consumer markets, understanding these interpretations early, is necessary for safe and confident

adoption of these vehicles. Our study was guided by the following research questions:

(1) **RQ1**: How do different forms of SAV communication (e.g., visual, auditory, haptic) shape user trust, cognitive load, and emotional comfort? We asked this question because communication is one of the few channels through which users can infer or have an idea of an SAV's internal state. Our findings show that these modalities do more than deliver information; they regulate how calm, prepared, or overwhelmed users feel. Thus, investigating the relationship between communication and trust, cognitive load and emotional comfort, helps reveal which modalities support understanding and which may unintentionally undermine trust [6, 8].
(2) **RQ2**: How do users negotiate control between themselves and the autonomous system during routine and emergency driving scenarios? Control is central to user confidence in SAVs and ultimately influences adoption of these vehicles, yet the transition between manual and automated modes is often where misunderstandings and safety failures happen. Exploring how users experience and desire control [5] especially when the SAV must intervene or when the user wants to override automation, helps clarify what kinds of shared-control models support safety and reduce anxiety.

Together, these research questions aim to explain not only what users think about SAVs but how they develop working relationships with them. These research questions evolved iteratively during analysis as themes emerged related to competence, control, and communication. Our key findings reveal that users judge SAV competence through comparisons to human limitations. We also see that they conceptualize driving as a negotiated partnership between human and machine, and evaluate communication modalities based on cognitive and emotional fit. The contribution of our work is a clearer understanding of the processes through which everyday users form trust and confidence, negotiate control boundaries, & align system communication with their expectations of safety and usability.

2 Method

2.1 Study Design and Data Collection

This study used a qualitative interview approach, specifically a reflexive thematic analysis approach, which emphasizes iterative coding, researcher subjectivity, and analytic flexibility [3] to explore how everyday users interact with and make sense of semi-autonomous vehicle systems. Two participants (AVU1 and AVU2) were recruited through convenience sampling. Both regularly operate a Tesla vehicle equipped with autonomous features such as autosteer, adaptive cruise control, and automated braking. We collected data through semi-structured interviews conducted via Zoom, each lasting approximately 30 minutes. The interview protocol included questions about perceptions of SAV communication, trust, system control, safety features, and past driving experiences. Also, the participants were encouraged to describe specific emergency scenarios where the SAV intervened as well as instances where they needed to reclaim control, or where system communication influenced their behavior. The interviews were recorded, transcribed verbatim, and then verified for accuracy, and no identifying information was retained.

2.2 Data Analysis

The analysis followed an inductive thematic approach informed by Braun and Clarke's reflexive thematic analysis framework. Our approach started with open coding, where we generated initial codes tied closely to participants' language. These codes were refined iteratively into more focused, analytic codes that captured the underlying patterns that were emerging from across the data. These codes were then compared, clustered, and interpreted to generate higher-level themes that connected user experiences across different contexts. The development of our themes emphasized analytic interpretation over surface description, particularly around how competence, control, and communication were conceptualized by the participants.

2.3 Researcher Positionality

As the researcher conducting and analyzing the interviews, my positionality shaped how I interpreted participants' experiences. My academic background in human-centered computing and my ongoing work in accessibility interaction likely positioned me to attend closely to issues of trust, transparency, and control dynamics because my experiences influence me to give more attention to how automation systems can be made safer and more trustworthy. Additionally, I also approached SAVs with a generally positive orientation toward their competence in superseeding human capabilities and their potential safety benefits, which may sensitize me to situations in which participants describe competence or predictability. Thus, I ensured to continually center participants' lived experiences rather than my own technological assumptions. Recognizing my own beliefs and expectations helped me inductively analyze the data without personal beliefs influencing my analysis. The full semi-structured interview guide is provided in the Appendix.

2.4 Limitations

While this study offers meaningful insight into early experiences with autonomous vehicle communication and control, several limitations influence the interpretation of the findings. Firstly, the sample size was very small, consisting of two participants, which limits the diversity of experiences represented. The goal of this study was depth rather than breadth, yet the views presented here cannot be generalized to all AV users or to populations with different driving habits, physical abilities, or familiarity with automation. In addition, participants were individuals who already had experience with consumer AV or advanced driver-assistance features. Their comfort with technology may differ significantly from users who are skeptical of automation or who have never interacted with AV systems. As a result, the themes identified here may overrepresent the perspectives of relatively tech-confident drivers. Also, the study relied on semi-structured interviews conducted over Zoom. While this method allowed for in-depth reflection, it did not include direct observation that often reveal behaviors, emotional reactions, and real-time breakdowns. Participants' accounts are therefore based on their remembrance of past events and may not fully capture live dynamics of collaboration with an AV system. In the same vein,

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because the interviews were not paired with real AV scenarios or simulations, participants often imagined hypothetical situations involving emergency intervention or communication breakdowns. These hypothetical reflections are valuable for getting expectations, but they may not accurately reflect how users behave under actual stress or uncertainty. Finally, the researcher's own positionality shapes the analysis. As someone trained in human-centered computing and familiar with AV design discourse, we may have interpreted participant comments through a lens that emphasizes system transparency, shared control, and communication modalities. While reflexive practices were used to mitigate this influence, the interpretive nature of qualitative analysis means that findings emerge through the interaction between participant accounts and researcher perspective. Together, these limitations do not diminish the value of the insights from this study, but rather clarify the scope of the contribution. They also point toward opportunities for future work, including larger and more diverse samples, scenario-based or simulation-based studies, and comparative analyses of users with varying levels of AV familiarity.

3 Findings

Through the qualitative analysis of interviews, we surfaced the lived experiences of the multiple stakeholders who interacted with autonomous vehicles. In this section, we present the findings of the study.

3.1 Users Judge SAV Competence Through Its Ability to Detect and Act Faster Than a Human

Participants consistently evaluated the SAV's value through its ability to perceive hazards more effectively than they could themselves. They frame the SAV's sensor and camera systems as offering expanded perception beyond their human capabilities. This expanded awareness functions as a core argument for adopting autonomous features, because the system can detect hazards the human cannot see. Rather than viewing autonomous driving as relinquishing of control, they described a sense of improved capabilities, which comes from what the vehicle can "see" and detect before the driver has the ability to do so. As AVU1 noted, driving in low-visibility nighttime environments with wildlife risks makes them actively prefer automation:

Participant quotation omitted in this public copy.

These findings closely reflect broader research in human–automation trust [6], which shows that users tend to rely on systems when they demonstrate abilities that clearly exceed human perception or response time. In our data, this comparison between human and machine capability was grounded in specific events in which the AV detected hazards way earlier than the participant could. This suggests that trust in SAVs is primarily based on competence and rooted in observable system behavior rather than general attitudes toward technology. This comparison of human and machine-supported perception also shows how participants conceptualize the SAV's perception as an extension of their own sensory limitations. The emphasis on improved perception also appeared during high-speed or complex traffic scenarios. AVU2 recalled a moment when the AV identified a fast-approaching car in the blind spot:

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These findings show that participants do not view AV adoption simply as a matter of convenience; rather, they describe the system as a safeguard whose perception capabilities surpass their own. This perceived "superhuman" awareness becomes the foundation for users' evaluations of whether the autonomous system is worth using at all.

Participant account and related discussion omitted in this public copy.

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Table 1: Codebook with Definitions and Supporting Quotes

Code	Definition	Supporting Quote
Predictive Sensing as Safety Advantage	Users describe AV sensors and cameras as extending human perception and detecting hazards earlier.	Participant quotations omitted in this public copy.
Emergency Intervention as Rationale for Adoption	Users rely on AVs to mitigate risks during health or safety emergencies.	
Immediate System Takeover as Safety Requirement	Interventions must happen instantly during emergencies to prevent harm.	
Desire for Override Mechanisms	Users want fast, simple ways to regain manual control.	
Visual Displays for Situational Awareness	Visual modality is the most trusted source of information about AV behavior.	
Context-Dependent Haptic/Audio Alerts	Audio and haptic cues are used selectively, mainly for urgency.	
Asynchronous Status Updates as Reassurance	Notifications outside the vehicle extend monitoring and reduce uncertainty.	
Onboarding Needed to Build Trust	Understanding features and safety systems increases early trust.	

This all goes to show that "trust" in the SAV's competence does not arise abstractly. Instead, users form judgments about whether an SAV earns their confidence based on perceived superhuman sensing and fast reaction times as well as understanding the features of the SAV. Trust is conditional, not assumed, and emerges when the SAV demonstrates capabilities users see as surpassing human abilities, and they understand the functioning of the system. Taken together, these findings suggest that competence and transparency work together as dual conditions for early trust. Participants were willing to credit the SAV with "superhuman" sensing capabilities, but this trust was reduced when they lacked understanding of how these capabilities functioned. Trust emerged most clearly when the SAV demonstrated superior perception in real-world events, and the participant had clarity about what the system was doing and why. This indicates that trust in SAVs is neither purely just performance-driven nor is it purely based on available information; rather, it is at the intersection of perceived capability and intelligibility.

3.2 Control of the Vehicle Is Negotiated Between the Human and the Machine

Participants discussed negotiation of control and the need for dynamic modes of gaining back control after AV intervention which could increase the sense of control that they have in the SAV. In discussing system responsiveness, they emphasized that emergency intervention must be instant because delays could worsen danger. In our data, participants did not see the SAV as a replacement driver but as a partner whose authority should change depending on risk, environmental complexity, and personal readiness. This shows their expectations for how quickly and seamlessly control should shift during both routine and emergency scenarios. When the participants were asked their preference for how the SAV would take control from them if it was in manual mode and they were in an emergency scenario, their preference was:

Participant quotation omitted in this public copy.

This preference for immediacy highlights the psychological dynamics of relinquishing control. Participants were willing to allow

automation to intervene only if the system demonstrated responsiveness that matched or exceeded their own reflexes. While participants valued the SAV's ability to intervene during safety-critical events, they also expressed a strong desire to retain the ability to reassert control at any moment. So, while they wanted the vehicle to be able to take control and switch from manual mode to auto mode, in the event that there was an emrgency scenario, they also wanted to be able to gain control of the SAv and switch from auto mode to manual mode. This functionality being available, especially in moments of need like the need to maneuver out of heavy traffic, could increase the user's sense of control by being able to commandeer the AV anytime they want.

Participant quotation omitted in this public copy.

While they expected instant machine action in emergency scenarios without a warning or confirmation prompt, yet in everyday driving, they wanted equally fast ways to override SAV automation. AVU2 described the ideal mechanism:

Participant quotation omitted in this public copy.

This need for a rapid override mechanism supports a conceptualization of driving as a shared control model in which authority moves fluidly between human and the machine. This balance of control is also influenced by context. AVU1 noted that when there are no incidents or need for intervention, they prefer to have control over putting the SAV in auto or manual mode, but they relinquish control strategically when environmental conditions reduce their driving confidence.

Participant quotation omitted in this public copy.

Thus, control becomes a negotiated and situation-dependent partnership rather than a fixed mode of operation.

3.3 Communication Modality Shapes Cognitive Load, Emotional Comfort, and Perceived System Transparency

Participants evaluated communication modalities not only as interfaces but as mechanisms shaping emotional comfort and cognitive load. Visual displays were preferred over other modalities like voice because they provide transparency and a real-time depiction of what the AV "sees," which reduces uncertainty supports user confidence. Participants rely on visual displays to maintain cognitive and spatial orientation during driving. AVU1 described visual displays as "my security," explaining that seeing the environment rendered on screen confirmed that the AV was perceiving hazards correctly.

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Audio and haptic notifications were less preferred in comparison to visual displays:.

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Audio and haptic notifications were valued for urgent situations but described as irritating or burdensome when used frequently. Participants wanted the communication modality to match the situational context. Thus beeps and haptics were preferred for emergencies, and visual cues for routine navigation. This fine-grained distinction suggests that communication is not merely informational but deeply tied to how users maintain situational awareness and regulate attention.

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Communication outside the vehicle was also discussed. AVU2 emphasized that asynchronous notifications provided reassurance by extending awareness beyond the driving context. Receiving push notifications about system status and environmental events made them "feel the best," reinforcing perceptions of the AV as a continuously monitoring partner.

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To also reduce anxiety and uncertainty when operating the AV for the first time, participants discussed the need for a structured, guided explanations of AV features.

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4 Discussion

The findings from this study show that early experiences with autonomous vehicles are influenced by three interdependent processes: evaluating system competence, negotiating shared control, and interpreting communication modalities. Together, these processes show how users build trust, develop confidence, and construct a mental model of the AV as a collaborative partner rather

than a replacement driver. In this section, we interpret these findings, relate them to broader research in human–automation interaction, and outline the implications for AV design, policy, and future deployment.

4.1 Perceived System Competence and the Conditions of Trust

The way users assess SAV competence aligns with prior research showing that trust in automation grows when systems reliably perform beyond human limitations, especially in detection and response time. Participants relied on the SAV's enhanced sensing capabilities to identify risks they themselves might miss. This perception of "superhuman awareness" directly shaped their trust and willingness to accept autonomous features. However, the findings are nuanced and don't just convey simple narratives of trust. Participants did not trust the SAV because it had advanced capabilities; rather, trust was conditional and they still required system transparency. Also, without structured onboarding, participants expressed uncertainty about what the SAV could and could not do. This insight contributes to the literature by highlighting trust as a knowledge-dependent construct. Participants wanted explanations, demonstrations, and clear descriptions of safety features before fully embracing autonomous functionality. This suggests that manufacturers and policymakers should treat onboarding as a safety component rather than an optional convenience. This is necessary as users might not be motivated to use SAVs if they lack clear, guided exposure to system capabilities. Structured onboarding, demonstrations, and transparent feature explanations could improve both safety and consumer confidence.

4.2 Shared Control as a Preferred Model of Human–Automation Interaction

Our findings on control preferences are supported by broader research on shared autonomy, where users prefer a partnership rather than full automation. Participants consistently described wanting flexible control boundaries that shift according to context such as dense traffic, emotional stress, perceived risk, or unexpected events. This introduces a nuanced contribution: users do not simply want "control" or "automation," but the ability to be able to dynamically negotiate between both of them. Immediate override mechanisms were seen as important, not optional. This suggests that SAVs designed around non-instant handover protocols may unintentionally impact user confidence, especially in safety-critical moments. The findings also raise important considerations for policymakers. If users expect immediate manual override, regulatory frameworks must address how SAVs will balance their safety logic with human authority. For example, should a human be allowed to override an automated emergency maneuver if they believe they can act faster? These questions have implications for liability, training requirements, and consumer expectations. Thus, future AV systems must support dynamic, shared control that is sensitive to situations, rather than the binary modes of manual vs. autonomous operation.

4.3 Communication Modalities as Emotional and Cognitive Regulators

Communication modality emerged as a main factor in shaping emotional comfort, situational awareness, and cognitive load. Participants preferred visual feedback because it allowed them to build a mental model of the SAV's reasoning. Audio and haptic cues were useful only when conveying urgency and overuse of these cues caused irritation or anxiety. This finding expands prior work by showing that multimodal communication is not simply an interface preference but an emotional and psychological regulator. The SAV's communication style influences if users interpret system behavior as predictable, competent, and aligned with their needs or not. Poorly calibrated communication risks undermining trust even if the underlying automation is functioning correctly. These insights have direct implications for interaction design. AVs should adapt communication modality to situational context like reserving audio alerts for urgent events but relying on visual displays for routine information. They should also account for user state (stress, distraction level, experience) to avoid unnecessary cognitive burden.

4.4 Broader Implications for Design, Training, Regulation, and Deployment

Taken together, the findings point toward a model of SAV adoption grounded not in blind trust but in supported confidence. Several implications emerge; **For Designers**: SAVs must explain their "reasoning" through visual, comprehensible displays. Onboarding guides should also be designed to walk users through safety features, limitations, and best practices. Also, override mechanisms must be physically accessible, intuitive, and responsive. **For Older Adults and New Users**: the provision of structured onboarding reduces anxiety and promotes safer early use. Transparent communication also helps to build stable expectations of SAV behavior. **For Industry**: SAV manufacturers may benefit from standardized onboarding protocols similar to aviation checklists or medical device safety training. **For Policy Regulators**: Regulators could require documentation or demonstration of safety features as part of AV delivery. Liability frameworks may also be encouraged to account for dynamic shared control, not simple manual vs. automated modes. Overall, the findings reinforce that SAVs must be designed not simply as technical systems but as collaborative partners that communicate clearly, act predictably, and support the human in maintaining situational and emotional control. As SAVs continue to enter everyday mobility, these insights can inform more accessible, trustworthy, and user-centered deployment strategies.

A Interview Protocol

A.1 Study Overview

This semi-structured interview study explores how users experience communication and collaboration with autonomous vehicles (AVs). The goal is to understand how different communication modalities (visual, auditory, and haptic) shape trust, cognitive load, and sense of control during both routine and emergency scenarios.

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A.2 Eligibility Criteria

- Adults aged 18 or older.
- Prior experience with any form of vehicle automation (e.g., ADAS features, Tesla Autopilot).

A.3 Data Collection Procedure

Interviews were conducted over Zoom and lasted approximately 30–45 minutes. Participants were informed of confidentiality and voluntary participation. No identifying information was collected. Questions were asked in a flexible order, and probing questions were used when appropriate to elicit richer detail.

A.4 Interview Guide

1. Experience and Context

- Can you tell me about your experience driving or riding in an autonomous vehicle?
- Probe: What excites or concerns you most about AVs?

2. Information Exchange

- When you interact with the AV, what kinds of information does it communicate to you, and how does it do so?
- Probe: Are these messages visual, spoken, or tactile?

3. Preferred Communication Modes

- What forms of communication (voice, screens, sounds, vibrations) make you feel most comfortable or confident?

4. Trust in Communication

- Can you describe a situation where the AV's communication made you feel trusting or distrusting of its decisions?
- Probe: What influenced your level of trust?

5. Sense of Control

- Have you ever felt less in control while using the AV? What contributed to that feeling?

6. Control Transition Preferences

- In what types of driving situations would you want to take control from the AV?
- Probe: How should the AV notify you that control is being transferred (sound, visual cue, vibration)?

7. Emergency Scenarios

- In an emergency, would you prefer the AV to remain in control or for you to take over? Why?
- Follow-up: What type of alert or signal would help you respond effectively?

8. Timing of Handover

- Would you want the control transfer to happen instantly or with a countdown? How much warning feels comfortable?

9. Accessibility and Personal Preferences

- Do certain communication modes (voice, visuals, haptics) feel more or less accessible or comfortable for you? Why?

10. Design Recommendations

- If you could give AV designers one piece of advice for improving communication and collaboration, what would it be?

References

[1] Alqahtani, T. Recent trends in the public acceptance of autonomous vehicles: A review. *Vehicles 7*, 2 (2025), 45.

[2] Bjørner, T. A priori user acceptance and the perceived driving pleasure in semi-autonomous and autonomous vehicles.

[3] Braun, V., and Clarke, V. Using thematic analysis in psychology. *Qualitative research in psychology 3*, 2 (2006), 77–101.

[4] Endsley, M. R. Autonomous driving systems: A preliminary naturalistic study of the tesla model s. *Journal of Cognitive Engineering and Decision Making 11*, 3 (2017), 225–238.

[5] Flemisch, F., Heesen, M., Hesse, T., Kelsch, J., Schieben, A., and Beller, J. Towards a dynamic balance between humans and automation: authority, ability, responsibility and control in shared and cooperative control situations. *Cognition, Technology & Work 14*, 1 (2012), 3–18.

[6] Lee, J. D., and See, K. A. Trust in automation: Designing for appropriate reliance. *Human factors 46*, 1 (2004), 50–80.

[7] Subramanian, H. V. *Supporting Human-AI Interaction from User Expectations to Mental Models.* PhD thesis, Missouri University of Science and Technology, 2025.

[8] Waytz, A., Heafner, J., and Epley, N. The mind in the machine: Anthropomorphism increases trust in an autonomous vehicle. *Journal of experimental social psychology 52* (2014), 113–117.